1. **Origin of replication (ori):** This is a sequence from where replication starts and piece of DNA when linked to this sequence can be made to replicate within the host cells.

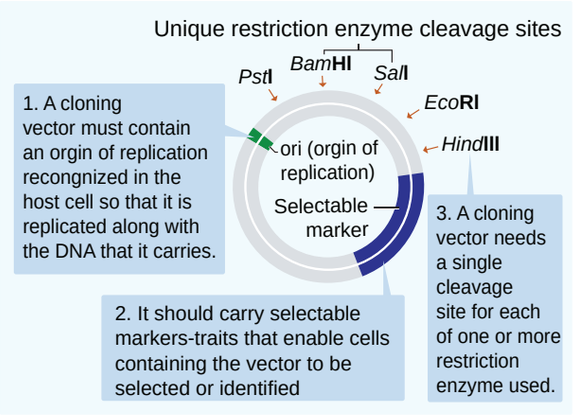


Figure 4.8: Properties of Vector

2. **Selectable marker**: In addition to **ori** the vector requires a selectable marker, which helps in identifying and eliminating non transformants and selectively permitting the growth of the transformants.
3. **Cloning sites**: In order to link the alien DNA, the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes.

Types of vector

Few types of vectors are discussed in detail below:

Plasmid

Plasmids are extra chromosomal, self replicating ds circular DNA molecules, found in the bacterial cells in addition to the bacterial chromosome. Plasmids contain Genetic information for their own replication.

pBR 322 Plasmid

pBR 322 plasmid is a reconstructed plasmid and most widely used as cloning vector; it contains 4361 base pairs. In pBR, ***p*** denotes plasmid, ***B***and ***R*** respectively the names of scientist **B**oliver and **R**odriguez who developed this plasmid. The number **322** is the number of plasmid developed from their laboratory. It contains amp^R and tet^R two different antibiotic resistance genes and recognition sites for several restriction enzymes. (*Hind III, EcoRI, BamH I, Sal I, Pvu II, Pst I, Cla I*), ori and antibiotic resistance genes. Rop codes for the proteins involved in the replication of the plasmid.

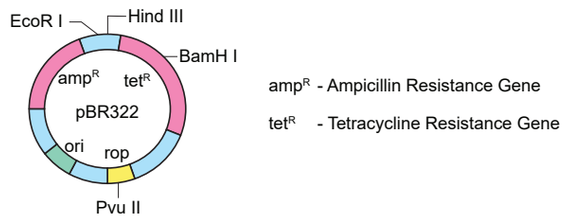


Figure 4.10: pBR 322

Ti Plasmid

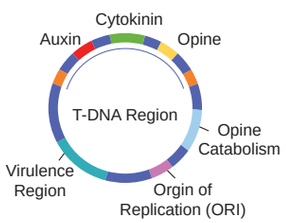


Figure 4.11: Ti Plasmid

Ti plasmid is found in *Agrobacterium tumefaciens,* a bacterium responsible for inducing tumours in several dicot plants. The plasmid carries transfer (tra) gene which help to transfer T- DNA from one bacterium to other bacterial or plant cell. It has Onc gene for oncogenecity, ori gene for origin for replication and inc gene for incompatibility. T-DNA of Ti-Plasmid is stably integrated with plant DNA. Agrobacterium plasmids have been used for introduction of genes of desirable traits into plants.

4.5.5 Competent Host (For Transformation with Recombinant DNA)

The propagation of the recombinant DNA molecules must occur inside a living system or host. Many types of host cells are available for gene cloning which includes *E.coli,* yeast, animal or plant cells. The type of host cell depends upon the cloning experiment. *E.coli* is the most widely used organism as its genetic make-up has been extensively studied, it is easy to handle and grow, can accept a range of vectors and has also been studied for safety. One more important feature of *E.coli* to be preferred as a host cell is that under optimal growing conditions the cells divide every 20 minutes.

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